

Voice Cloning Air Traffic Control: Vulnerabilities at Runway Crossings

Andrew Logan

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Never fear, I is here.





Let's rehearse that.



Who Am I

Andrew Logan

Professional Audio Engineer

Def Con 30 (2022): "Tracking
the Military Ghost Helicopters
of DC"



<https://www.congress.gov/members/find-your-member>

Say you support:

Safe Operation of Shared Airspace Act of 2025

Bills that broaden the use of ADS-B



My Nightmare:

- A sophisticated attacker impersonating air traffic controllers
- Intermittently giving false clearances to cross runways
- In adverse weather conditions
- Could cause crashes
- More broadly undermine the legitimacy of air traffic control



We've got you.

PULL UP!



FCC



WARNING

Federal law prohibits transmitting on airband frequencies.

Interfering with air traffic control is punishable by a \$10,000 fine and up to 10 years in prison .



01

**AM “Airband”
Radio**



01

**AM “Airband”
Radio**

02

**Voice Cloning
Technologies**



01

**AM “Airband”
Radio**

03

**Why runway
crossings are
most at risk**

02

**Voice Cloning
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**Voice Cloning
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04

**How to
mitigate this
threat**

DISCLOSURES

- **U.S. Senate Committee on Commerce, Science, & Transportation**
- **NATCA – National Air Traffic Controller Assoc.**
- ICAO – International Civil Aviation Organization
- CISA - Cybersecurity and Infrastructure Security Agency
- FAA – Federal Aviation Association
- FCC - Federal Communications Commission

PLAY ▶

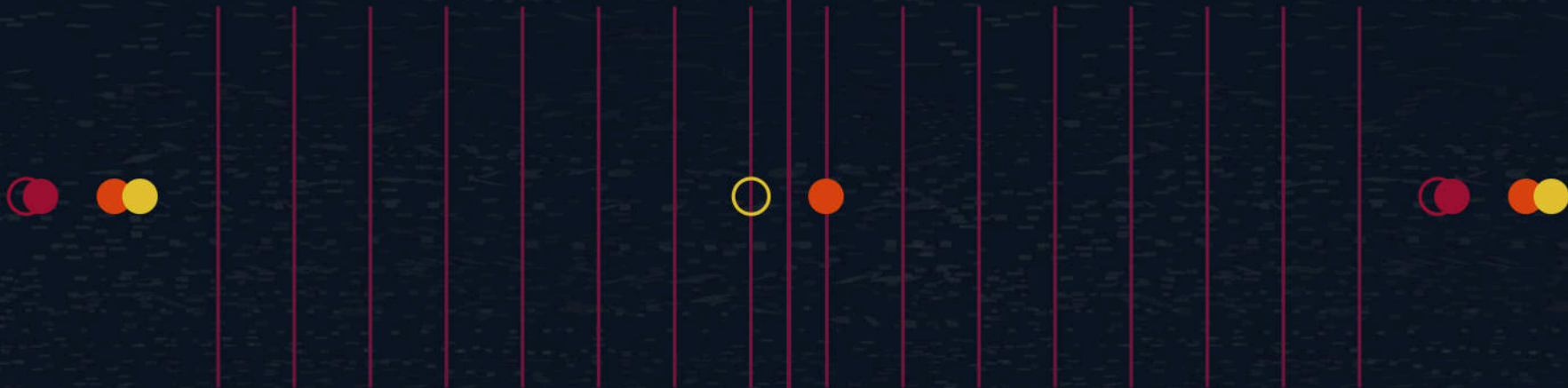
01

AM RADIO

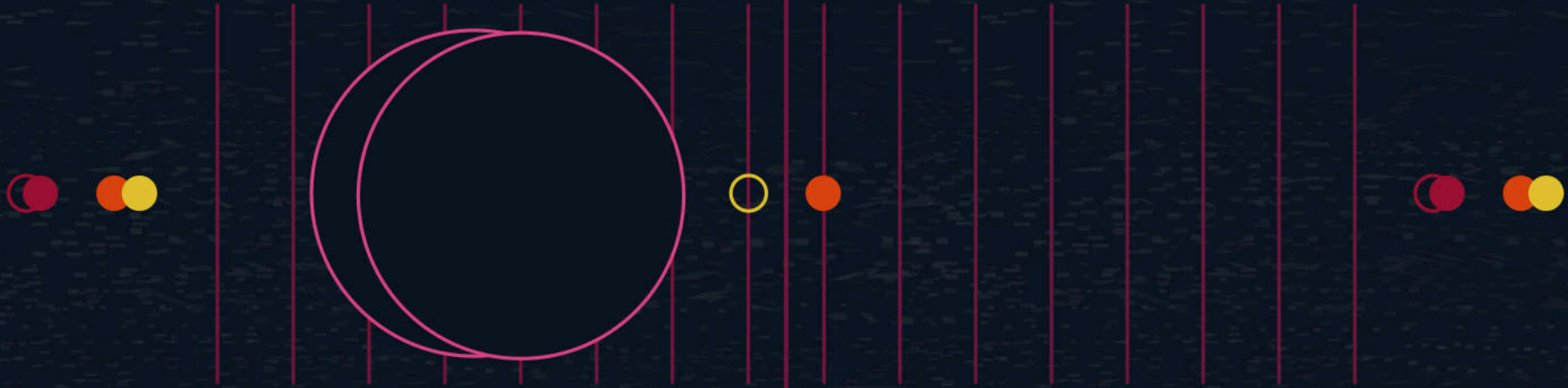


everything's perfectly all right now. We're fine. We're all fine here, now, thank you. How are you?

AM RADIO CHARACTERISTICS

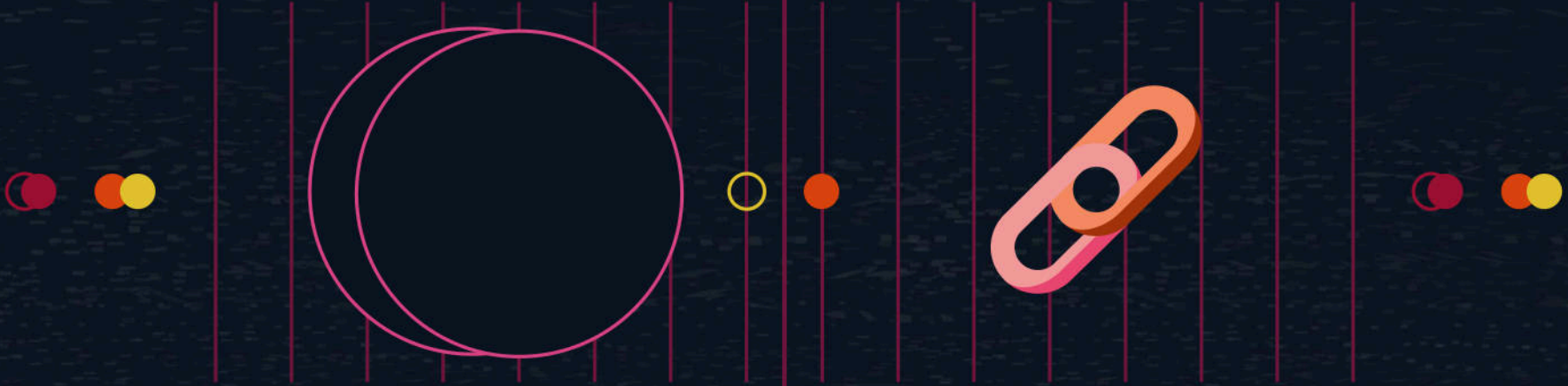


AM RADIO CHARACTERISTICS



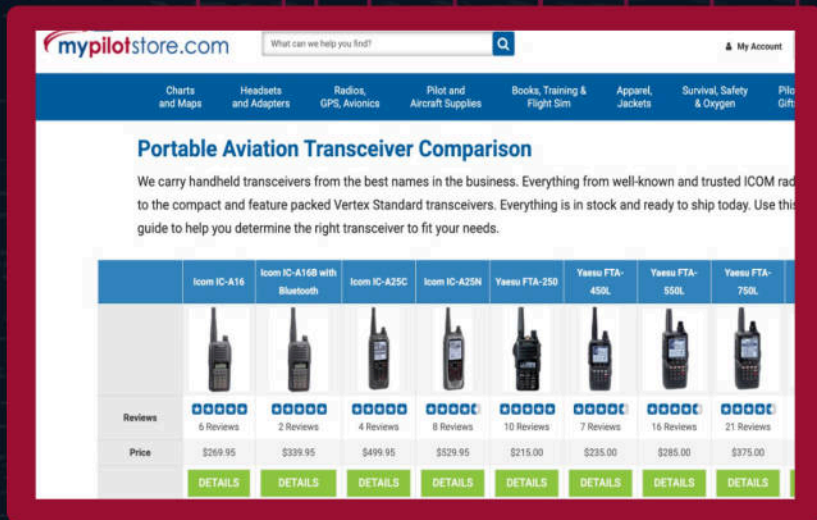
Allows for “heterodyning
of carriers”

AM RADIO CHARACTERISTICS



Allows for “heterodyning
of carriers”

Weak Capture effect
compared to FM and
digital modes



Portable Aviation Transceiver Comparison

We carry handheld transceivers from the best names in the business. Everything from well-known and trusted ICOM radios to the compact and feature packed Vertex Standard transceivers. Everything is in stock and ready to ship today. Use this guide to help you determine the right transceiver to fit your needs.

	Icom IC-A16	Icom IC-A16B with Bluetooth	Icom IC-A25C	Icom IC-A25N	Yaesu FTA-250	Yaesu FTA-450L	Yaesu FTA-550L	Yaesu FTA-750L
								
Reviews	 6 Reviews	 2 Reviews	 4 Reviews	 6 Reviews	 10 Reviews	 7 Reviews	 16 Reviews	 21 Reviews
Price	\$269.95	\$339.95	\$499.95	\$529.95	\$215.00	\$235.00	\$285.00	\$375.00
	DETAILS	DETAILS	DETAILS	DETAILS	DETAILS	DETAILS	DETAILS	DETAILS

**Commercially
Available AM
Transceivers**



Secondhand Tower Systems



Motorola CM-400 VT VHF Transmitter Air Band Make Offer!



Shems Electric LTD (490)

100% positive · [Seller's other items](#) · [Contact seller](#)



US \$400.00

or Best Offer

as low as \$37.05/mo with **Klarna**. [Learn more](#)

Condition: **For parts or not working** ⓘ

Buy It Now

- HF Radio (analog)

Digital and Other Modes

Digital and Other Modes

- HF Radio (analog)
- Aircraft Communications Addressing and Reporting System (ACARS)

Digital and Other Modes

- HF Radio (analog)
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- VHF Data Link Mode 2 (VDL Mode 2)

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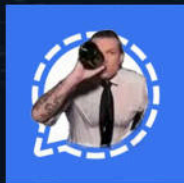
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INTERPERSONAL ELEMENTS OF ATC

is it a single factor trust system?

01

Channel

Are you transmitting on
the correct channel

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Voice

Do you sound authentic

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Syntax & Validity

Is the command formatted and timed correctly

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is it a single factor trust system?

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Voice

Do you sound authentic

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Syntax & Validity

Is the command formatted and timed correctly

04

Expectation

Is the command expected or a response to a prior request

FCC ENFORCEMENT TECH



Direction Finding Vehicles

Say s*** and f*** a million
times if you want to! Steal
the air!



FCC ENFORCEMENT TECH



**Direction Finding
Vehicles**



**Fixed Direction
Finding Stations**

Say s*** and f*** a million
times if you want to! Steal
the air!



FCC ENFORCEMENT TECH



**Direction Finding
Vehicles**



**Fixed Direction
Finding Stations**



**Field
Investigative
Agents**

Say s*** and f*** a million
times if you want to! Steal
the air!





Intermittent

VS



Persistent

The call is coming from
inside the house!



PLAY ▶

02

Voice Cloning

ROONEY!

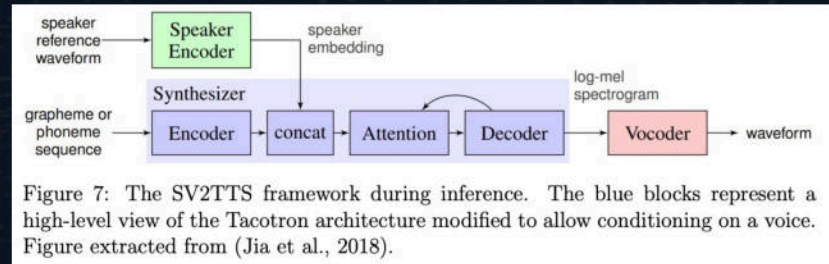


VOICE CLONING TECH

01

Voice Encoder

Captures Voice Characteristics



VOICE CLONING TECH

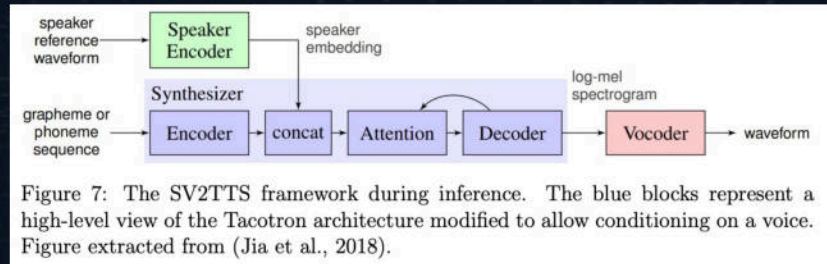
01

Voice Encoder

Captures Voice Characteristics

02

Text-to-spectrogram Synthesizer
generates a spectrogram from text



VOICE CLONING TECH

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Voice Encoder

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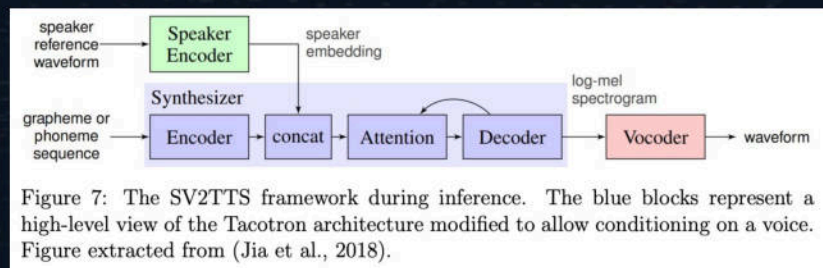
Text-to-spectrogram Synthesizer

generates a spectrogram from text

03

Vocoder

infers an audio waveform from the spectrogram



Guardrails

Guardrails

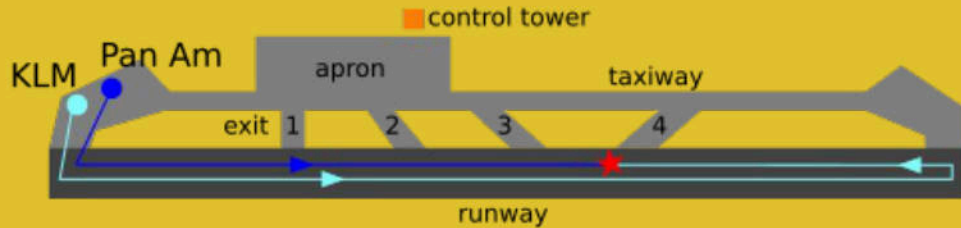
- Verification before cloning celebrities/public figures

PLAY▶

03

RUNWAY CROSSINGS

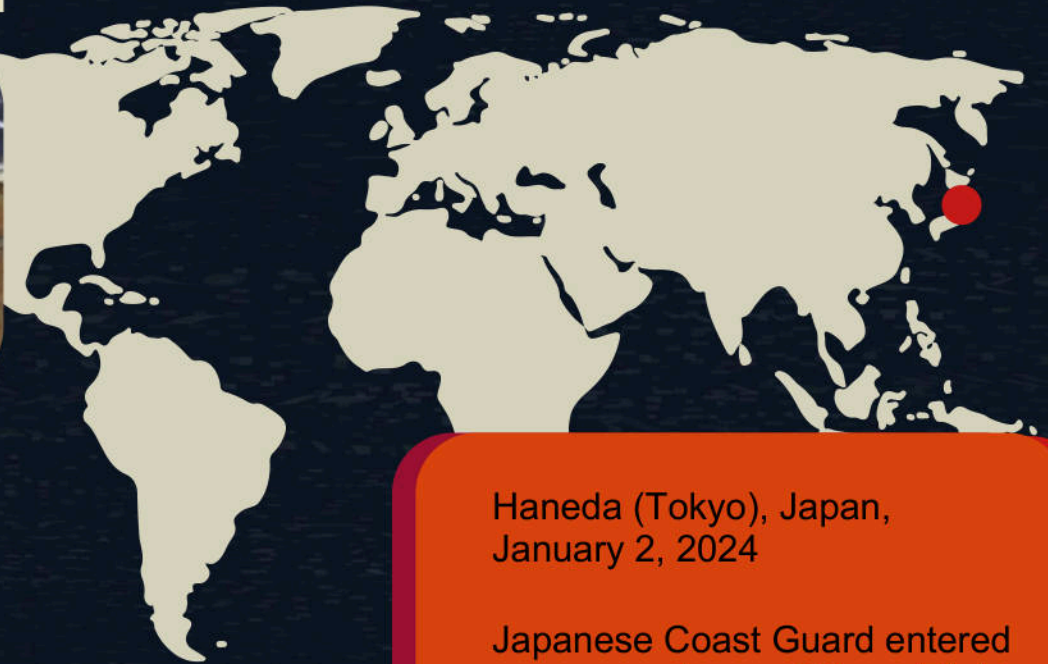
Tenerife Disaster



Los Rodeos Airport, **Tenerife**,
Canary Islands, Spain, March
27, 1977

583 fatalities: deadliest accident
in aviation history.

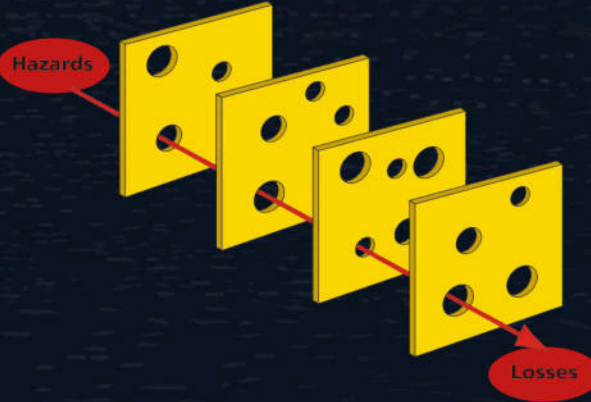
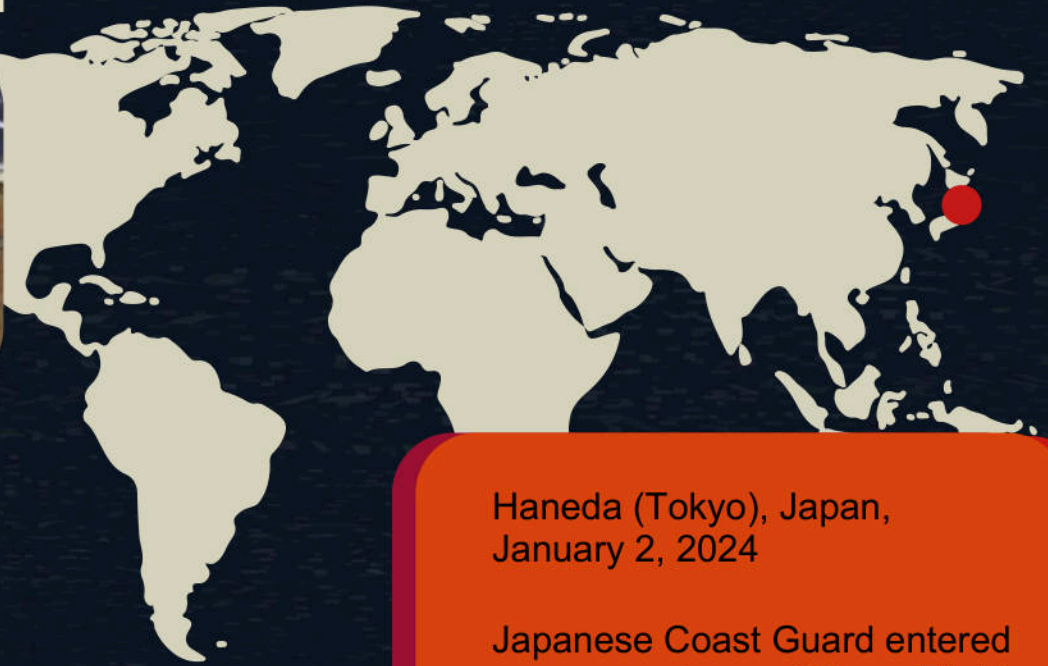
Haneda Crash



Haneda (Tokyo), Japan,
January 2, 2024

Japanese Coast Guard entered
runway as Japan Airlines Flight
516 was landing

Haneda Crash



Haneda (Tokyo), Japan,
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Japanese Coast Guard entered
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516 was landing

"Speed bird" (British Airways) 173A

22R

22L

BAW173A

Date: 25-07-02

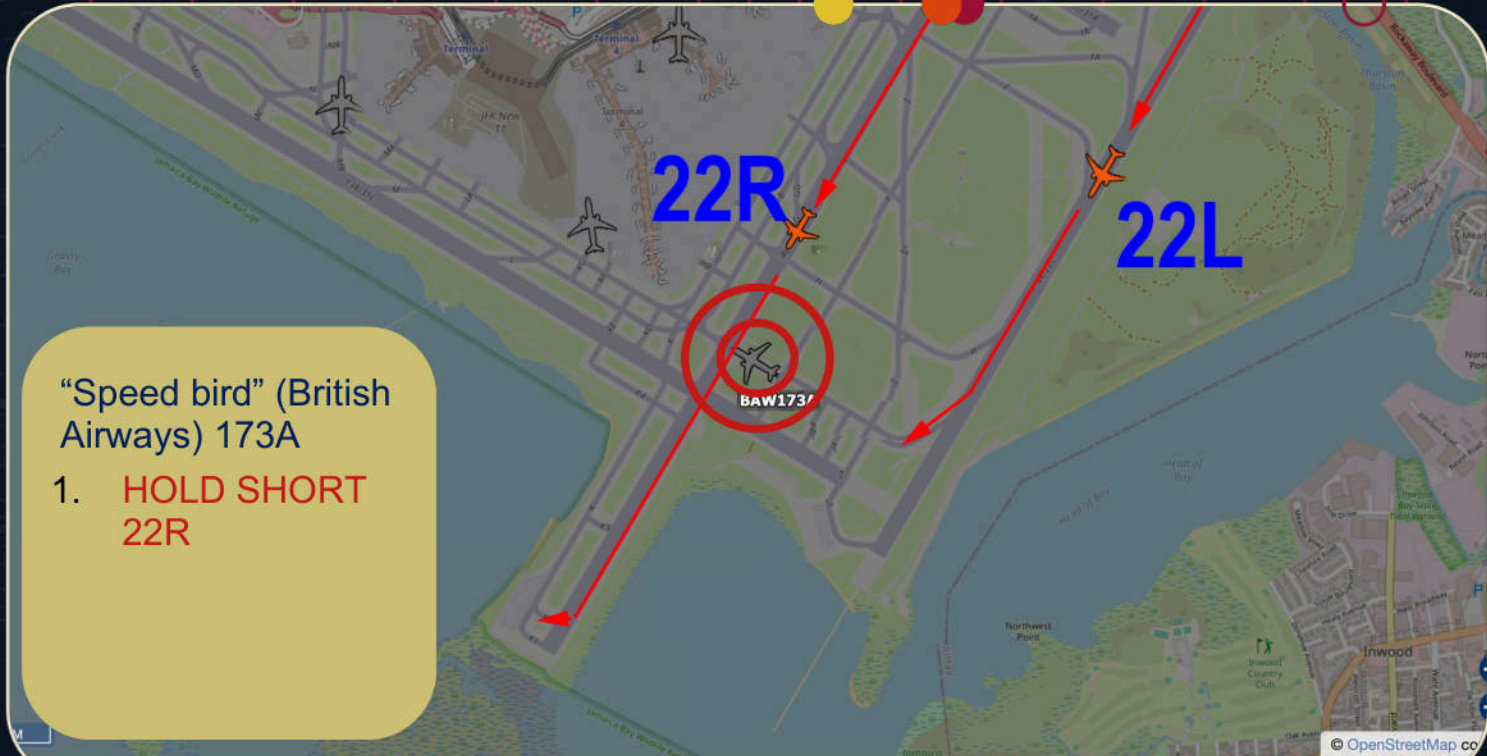
Time: 18:02:00 Z

Speed: 1.0x

© OpenStreetMap co

"Speed bird" (British Airways) 173A

1. **HOLD SHORT 22R**



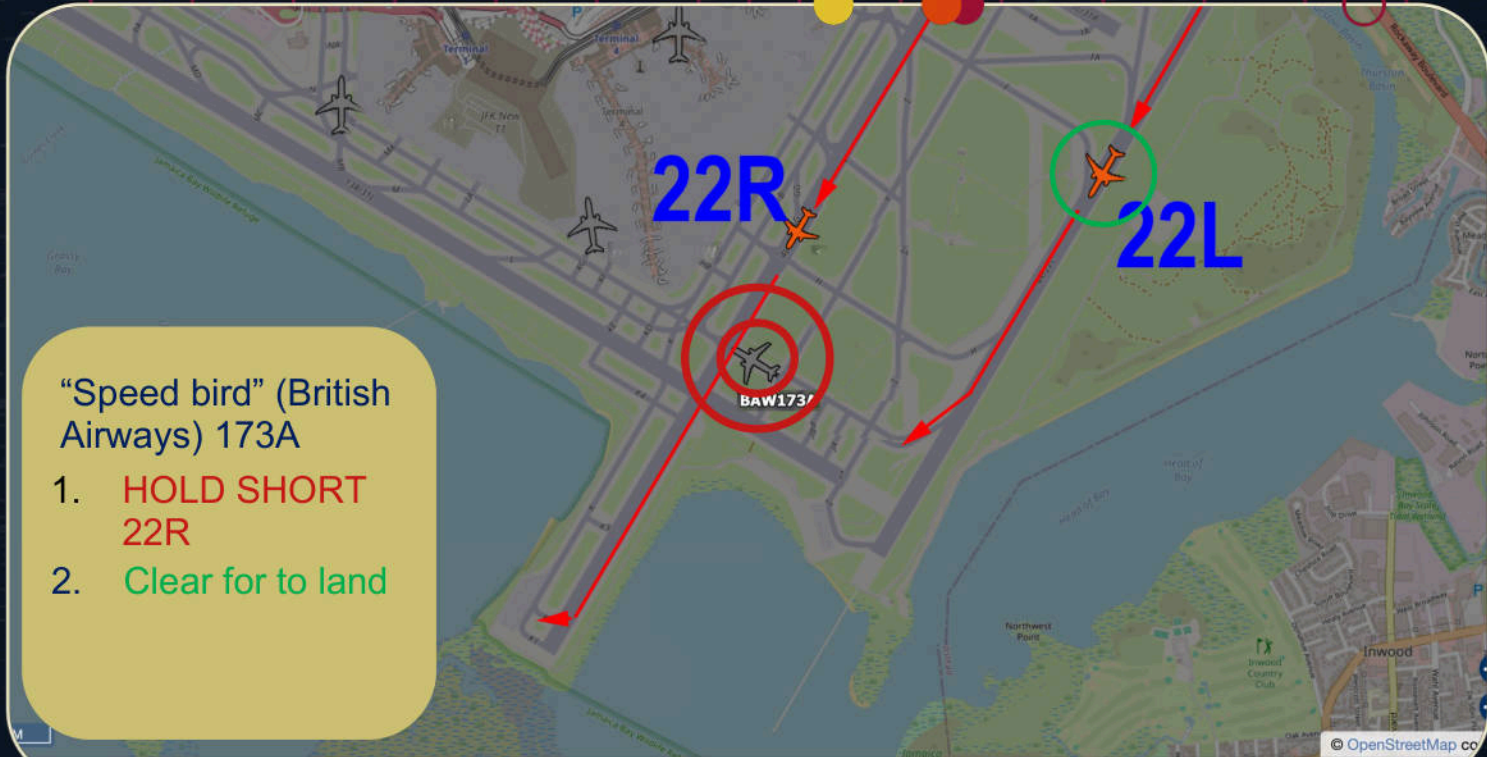
Date: 25-07-02

Time: 18:02:00 Z

Speed: 1.0x

"Speed bird" (British Airways 173A)

1. **HOLD SHORT**
22R
2. **Clear for to land**



Date: 25-07-02

Time: 18:02:00 Z

Speed: 1.0x

"Speed bird" (British Airways) 173A

1. **HOLD SHORT**
22R
2. **Clear for to land**
3. **[Cloned] Cross**
Runway

Date: 25-07-02

Time: 18:02:00 Z



Additional Audio Examples

Additional Audio Examples



PLAY ▶

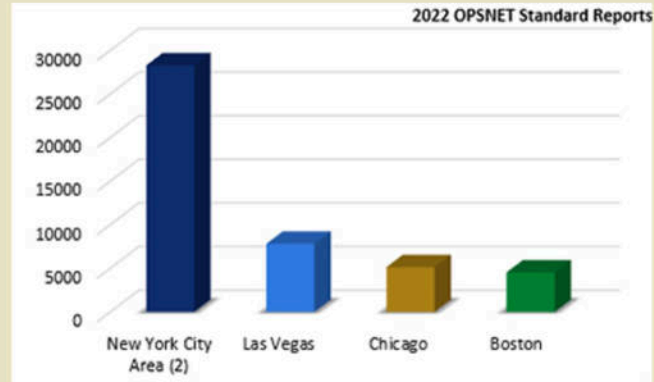
04

MITIGATION



Make a move and the bunny gets it.

Threat Modeling



FAA Airports with most weather delays

STOP BARS

"If the stop bar lights have not been switched off after receipt of a clearance, the pilot or vehicle driver **must** seek clarification from ATC. Do not cross an illuminated stop bar."



Automation and Enhanced Vision in the cockpit

- Traffic Collision Avoidance System (TCAS)
- Enhanced Flight Vision Systems
 - Cameras
 - Infrared cameras
 - Radar



Sources via [Google Sheet](#)

Andrew Logan

X/Instagram/BlueSky: [@HelicoptersofDC](#)

Mastodon: [HelicoptersofDC@DefCon.Social](#)

[HelicoptersofDC@ProtonMail.com](#)

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